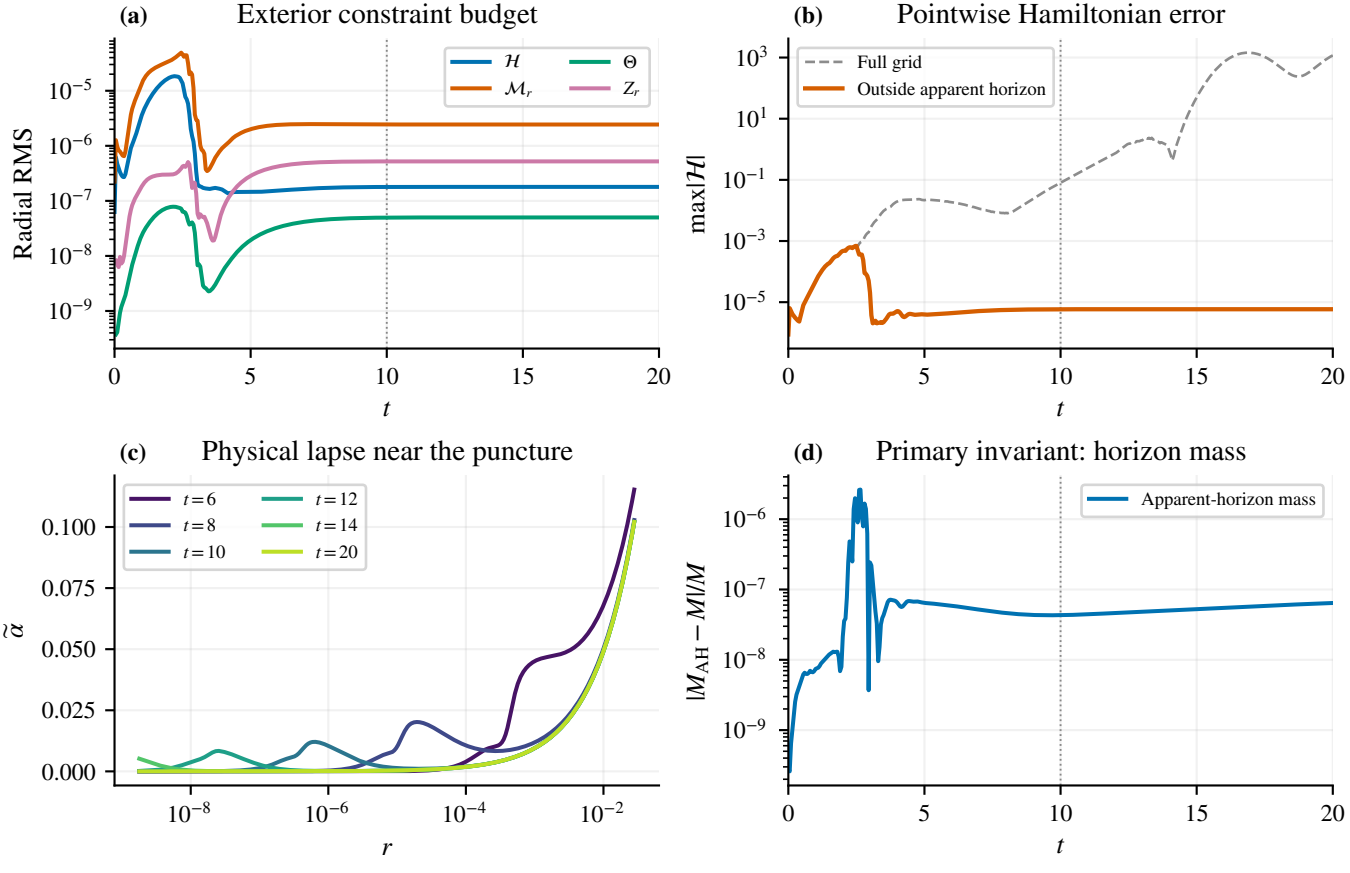
$N = 1025$ puncture pilot: target $t = 500$, evidence through $t = 20$
(a)
Exterior constraint budget
Radial RMS
$\mathcal{H}$
$\mathcal{M}_r$
Θ
Z_r
t
(b)
Pointwise Hamiltonian error
$\max|\mathcal{H}|$
Full grid
Outside apparent horizon
t
(c)
Physical lapse near the puncture
$\tilde{\alpha}$
$t = 6$
$t = 8$
$t = 10$
$t = 12$
$t = 14$
$t = 20$
r
(d)
Primary invariant: horizon mass
$|M_{\mathrm{AH}} - M|/M$
Apparent-horizon mass
t